GC University Lahore

EXAMINATION: FINAL 2025 **SEMESTER: VI SESSION (2022-26)**

COURSE CODE: CS-2235/CS-3102 **TITLE: ARTIFICIAL INTELLIGENCE**

Time Allowed: 25 minutes **Total Marks: 10**

Name: __________________ **Roll No: ___________**

OBJECTIVE PART

1. Which of the following best describes an **intelligent agent**?
   a) A human expert system
   b) A machine that performs a task once
   c) A system that perceives and acts in its environment
   d) A chatbot only

2. What was Alan Turing's major contribution to AI?
   a) Designed the first robot
   b) Developed a model of human learning
   c) Proposed the Turing Test for machine intelligence
   d) Created the first neural network

3. In reinforcement learning, an agent learns by:
   a) Memorizing examples
   b) Receiving rewards and penalties
   c) Predicting outputs directly
   d) Clustering data

4. A **neuron** in an artificial neural network consists of:
   a) Rules and clauses
   b) Dendrites, axons, and weights
   c) Input, weights, activation function
   d) Actuators and sensors

5. Which search algorithm explores nodes **level by level**?
   a) DFS
   b) BFS
   c) A*
   d) Best-First Search

6. In A* search, which formula is used to evaluate nodes?
   a) f(n) = g(n)
   b) f(n) = h(n)

c) $f(n) = g(n) + h(n)$
d) $f(n) = n + 1$

7. Bayesian networks help in:
   a) Image compression
   b) Probabilistic reasoning
   c) Sorting large data
   d) Neural activation

8. In a neural network, a **neuron** is modeled after:
   a) Logical circuits
   b) The human brain's nerve cells
   c) SQL databases
   d) Robotic arms

9. Which search algorithm expands the node that appears to be **closest to the goal**, based on heuristic only? a) A*
   b) Uniform Cost
   c) Greedy Best-First Search
   d) Iterative Deepening

10. Which algorithm performs **depth-limited search repeatedly**, increasing the limit each time?
    a) Greedy Search
    b) BFS
    c) Iterative Deepening DFS
    d) Hill Climbing

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COURSE CODE: CS-2235/CS-3102 **TITLE: ARTIFICIAL INTELLIGENCE**

Time Allowed: 155 minutes **Total Marks: 40**

Name: __________________ **Roll No: ___________**

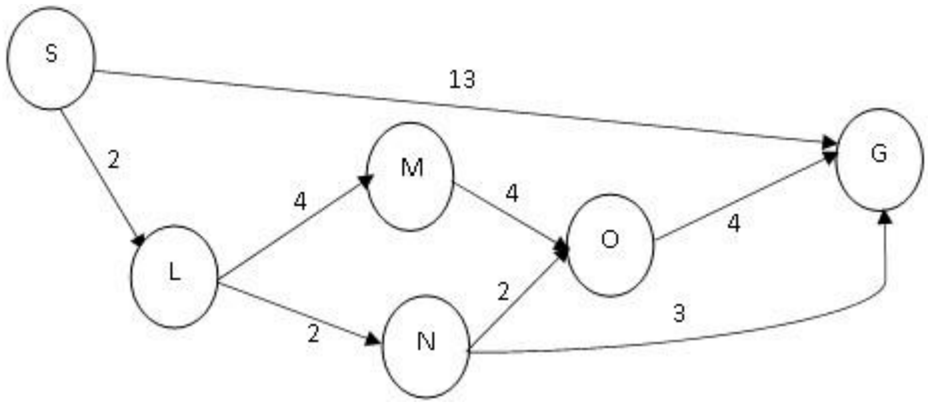
Each Question carries 10 marks. Attempt any 4 questions

Question No 1:

Given a binary tree (branching factor = 2), and a goal at depth 5:
a) Calculate the **total number of nodes visited** by DFS (without visited set).
b) Calculate the **total number of nodes visited** by BFS.
c) Compare memory usage and explain which is more space-efficient and why.

Question No 2:

Answer the questions in lieu of search problem given below.



1. What path would breadth-first graph search would return?
2. What path would depth-first graph search would return?
3. What path would uniform cost graph search would return?
4. What path would a* search would return?

STATE	**S**	**L**	**M**	**N**	**O**	**G**
HEURISTIC	5	3	4	2	1	0

5. Consider the following heuristics

STATE	**S**	**L**	**M**	**N**	**O**	**G**
H1	6	4	7	3	4	0
H2	5	3	7	1	4	0

a. IS H1 ADMISSIBLE AND CONSISTENT?
b. IS H2 ADMISSIBLE AND CONSISTENT?

Question No 3:

Consider the following 8-puzzle game:

Starting state (S)

Goal State (G)

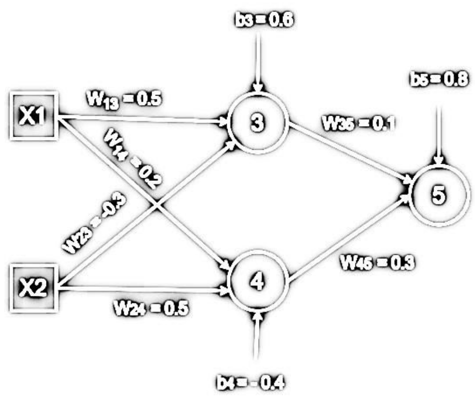
a. Make a three level search tree
b. Apply Hill Climbing to solve the problem. Use number of misplaced titles as heuristic value.
c. Does your search solve your problem in this case? Explain the reason

Question No 4:

Consider a *multilayer feed-forward neural network* given below. Let the *learning rate* be *0.5*. Assume initial values of weights and biases as given below. Train the network for the training tuples *(1, 1, 0)* where last number is target output.Show weight and bias updates by using *backpropagation algorithm*. Assume that *sigmoid activation function* is used in the network.

$$\text{(i.e. } \sigma = \frac{1}{1+ e^{-x}} \text{)}$$

W13	W14	W23	W24	W35	W45	b3	b4	b5
0.5	0.2	-0.3	0.5	0.1	0.3	0.6	-0.4	0.8



Question No 5:

a) Suppose there is a fitness function, `f(x) = [(3e + 5f + 7g + 9h) - 50]`. A genetic algorithm uses chromosomes of the genes, `e`, `f`, `g`, and `h`. The values of the four genes are integers between 10 and 30.

Let the initial population consists of six chromosomes with the value of gene `e`, `f`, `g`, and `h`, as follows:

X_1 = [11;25;23;18]
X_2 = [12;19;18;29]
X_3 = [10;14;13;24]
X_4 = [20;17;10;26]
X_5 = [18;24;13;19]
X_6 = [20;15;17;19]

i) Calculate the fitness and the fitness ratio of each individual chromosome and fill in the table below. Show your calculation.

Chromosome	Chromosome Fitness	Fitness Ratio, %
X_1		
X_2		
X_3		
X_4		
X_5		
X_6		

(1

ii) Arrange the chromosomes in ascending order.